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Equations Useful for Undergrad Physics Students

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Abstract

These are the Equations and Relations that i needed during studying in the Physics Departement as an undergrad student, I had problems memorizimg most of these.

CONTENTS

CHAPTER 1

VECTORS

$$\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(A.C) - \vec{C}(A.B) \tag{1.1}$$

Line Element in different co-ordinates:-

in Cartisian co-ordinates:

$$dl^2 = dx^2 + dy^2 + dz^2 \tag{1.2}$$

in Cylinderical co-ordinates:

$$dl^2 = dr^2 + r^2 d\phi^2 + dz^2 \tag{1.3}$$

in spherical Co-ordiantes

$$dl^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \tag{1.4}$$

CHAPTER 2

TRIGONOMETRY

$$Sin(\alpha \pm \beta) = Sin(\alpha)Cos(\beta) \pm Cos(\alpha)Sin(\beta) \tag{2.1}$$

$$Cos(\alpha \pm \beta) = Cos(\alpha)Cos(\beta) \mp Sin(\alpha)Sin(\beta) \tag{2.2}$$

$$Sin(\alpha)Cos(\beta) = \frac{1}{2}[Sin(\alpha + \beta) + Sin(\alpha - \beta)] \tag{2.3}$$

$$Cos(\alpha)Cos(\beta) = \frac{1}{2}[Cos(\alpha + \beta) + Cos(\alpha + \beta)] \tag{2.4}$$

$$Sin(\alpha)Sin(\beta) = \frac{1}{2}[Cos(\alpha - \beta) - Cos(\alpha + \beta)] \tag{2.5}$$

$$Sin^2\alpha = \frac{1}{2}[1 - Cos2\alpha] \tag{2.6}$$

$$Cos^2\alpha = \frac{1}{2}[1 + Cos2\alpha] \tag{2.7}$$

$$2SinxCosx = Sin2x \tag{2.8}$$

$$Sinx = \frac{e^{ix} - e^{-ix}}{2i} \tag{2.9}$$

$$Cosx = \frac{e^{ix} + e^{-ix}}{2} \tag{2.10}$$

$$\int_0^{\pi} Sin(\frac{n\pi}{a}x)Sin(\frac{n'\pi}{a}x)dx = \delta_{nn'} \tag{2.11}$$

Trignometric Differentiation

$$\frac{d}{dx}\ Sin\ x = Cos\ x \tag{2.12}$$

$$\frac{d}{dx}\ Cos\ x = -Sin\ x \tag{2.13}$$

$$\frac{d}{dx}\ Tan\ x = Sec^2\ x \tag{2.14}$$

$$\frac{d}{dx}\ Csc\ x = -Csc\ x\ Cot\ x \tag{2.15}$$

$$\frac{d}{dx}\ Sec\ x = Sec\ x\ Tan\ x \tag{2.16}$$

$$\frac{d}{dx}\ Cotx = -Csc^2\ x \tag{2.17}$$

CHAPTER 3

HYPERBOLIC FUNCTIONS

$$Sinhx = \frac{e^x - e^{-x}}{2} \tag{3.1}$$

$$Coshx = \frac{e^x + e^{-x}}{2} \tag{3.2}$$

$$Tanhx = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{3.3}$$

$$\frac{\partial}{\partial x}(Cosh(x)) = Sinh(x) \tag{3.4}$$

$$\frac{\partial}{\partial x}(Sinh(x)) = Cosh(x) \tag{3.5}$$

$$\frac{\partial}{\partial x}(Tanh(x)) = Sech^2(x) \tag{3.6}$$

$$\frac{\partial}{\partial x}(Sech(x)) = -Sech(x)Tanh(x) \tag{3.7}$$

$$\frac{\partial}{\partial x}(Csch(x)) = -Coth(x)Csch(x) \tag{3.8}$$

$$\frac{\partial}{\partial x}(Coth(x)) = -Csch^2(x) \tag{3.9}$$

CHAPTER 4

DIFFERENTIATION

$$y = \frac{u}{v} \qquad \longrightarrow \qquad \frac{dy}{dx} = \frac{vu' - uv'}{v^2} \tag{4.1}$$

Trignometric Differentiation

CHAPTER 5

EXPANTIONS

Theorem 1: Theorem Name

A theorem.

CHAPTER 6

FUNCTIONAL

$$\frac{\delta F}{\delta f(x)} = lim_{\epsilon \to 0} \frac{F[f(x) + \epsilon \delta(x - x')] - F[f(x)]}{\epsilon} \tag{6.1}$$

$$J[x] = \int f^p(x)\phi(x)\ dx \qquad \longrightarrow \frac{\delta\ J}{\delta\ f(x_o)} = P[f(x_o)]^{p-1}\phi(x_o) \tag{6.2}$$

$$H[f(x)] = \int g[f(x)]\ dx \quad \longrightarrow \frac{\delta H}{\delta f(x_o)} = g'(f(x_o)) \tag{6.3}$$

$$J[f] = \int g(f'(y))\ dy \qquad \longrightarrow \frac{\delta J}{\delta f(x_o)} = -\frac{1}{dx}(\frac{d\ g(f')}{d\ f'(x)}) \tag{6.4}$$

$$J[f] = \int (\frac{\partial f}{\partial y})^2\ dy \qquad \longrightarrow \frac{\delta J}{\delta f(x_o)} = -2\frac{\partial^2 f}{\partial y^2} \tag{6.5}$$

$$G[f] = \int g(y, f) dy \qquad \longrightarrow \frac{\delta H}{\delta f(x_o)} = \frac{\partial\ g(x, f)}{\partial f(x)} \tag{6.6}$$

$$J[f] = \int g(y, f, f', f'')\ dy \ \longrightarrow \frac{\delta J}{\delta f(x_o)} = \frac{\partial g}{\partial f} - \frac{d}{dx}(\frac{\partial\ g}{\partial f(x)}) + \frac{d^2}{dx^2}(\frac{\partial g}{\partial f''}) \tag{6.7}$$

CHAPTER 7

MAXWELL EQUATION

$$ \tag{7.1}$$

$$\nabla\times\text{E}=-\frac{\partial B}{\partial t} \tag{7.2}$$

$$\nabla.B = 0 \tag{7.3}$$

$$\nabla \times B = \mu_o J + \frac{1}{c^2}\frac{\partial E}{\partial t} \tag{7.4}$$

CHAPTER 8

GAMMA FUNCTIONS

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt \tag{8.1}$$

$$\Gamma(n) = (n-1)! \tag{8.2}$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi} \tag{8.3}$$

$$\int ln(x)\, dx = xln(x) - x \qquad lnN! = NlnN - N \tag{8.4}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}} \tag{8.5}$$

$$\int_{-\infty}^{\infty} x^n e^{-\alpha x^2} = \begin{cases} \frac{1}{2}(\frac{1}{\alpha})^{n+1} n! & n = 1, 2, 3..... \\ \frac{1}{2}(\frac{1}{\alpha})^{\frac{2n+1}{2}} \Gamma(n+\frac{1}{2}) & n = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, ... \end{cases} \tag{8.6}$$

CHAPTER 9

ESSENTIAL SERIES

$$\sum_{n=0}^{N-1} e^{inx} = \frac{1 - e^{iNx}}{1 - e^{ix}} \tag{9.1}$$